Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning using Power Query

Submitted By:

Name: Anuska Ghosh

Reg. No.: 2411021240012

Date: 11/09/26

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

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Data Cleaning using Power Query

GitHub Repository Link: *https://github.com/anuskaGHS/Business-Analytics*

Problem Statement

A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.
Dataset used: Cafe Sales Dirty Data

Source: *https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training*

Methodology / Steps Followed

Step 1: Data Collection

The Cafe Sales Dirty Data dataset was used for this experiment. The dataset was sourced from Kaggle and downloaded as a single CSV file. It contains cafe transactional data with the following key fields:

- Transaction ID, Item, Quantity
- Price Per Unit, Total Spent
- Payment Method, Location
- Transaction Date

Total Records: 10,000 rows
Total Columns: 8

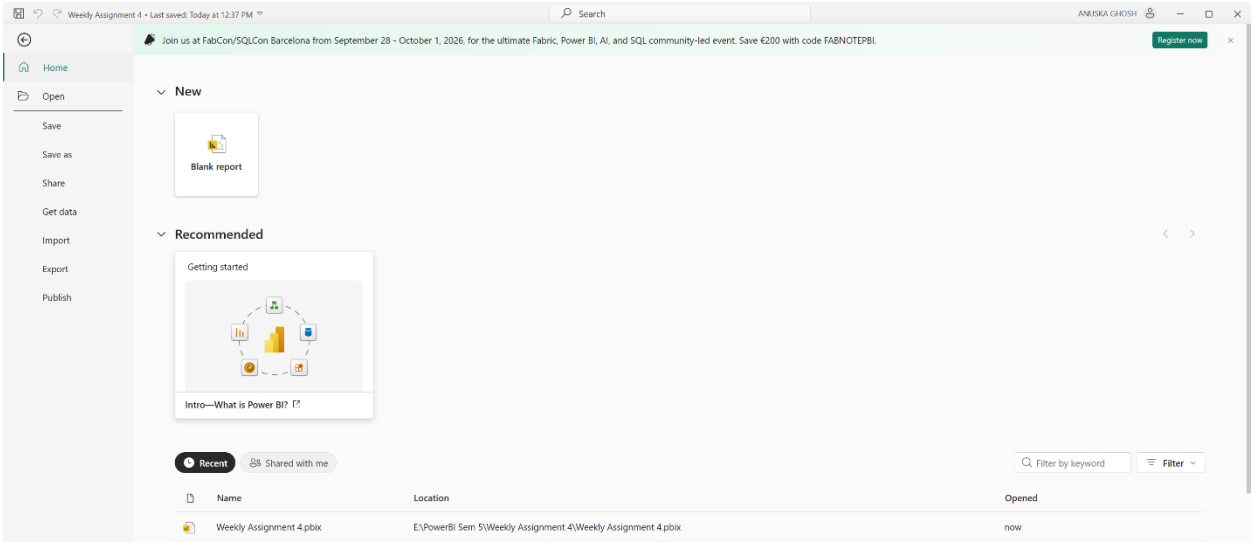


Figure 1: Power BI Desktop Home Screen

Step 2: Data Import in Power BI

Power BI Desktop was opened and the dataset was imported using the Get Data option by selecting the Text/CSV connector. The file dirty_cafe_sales.csv was loaded directly as a flat file. Power Query Editor was launched to inspect and clean the data before analysis.

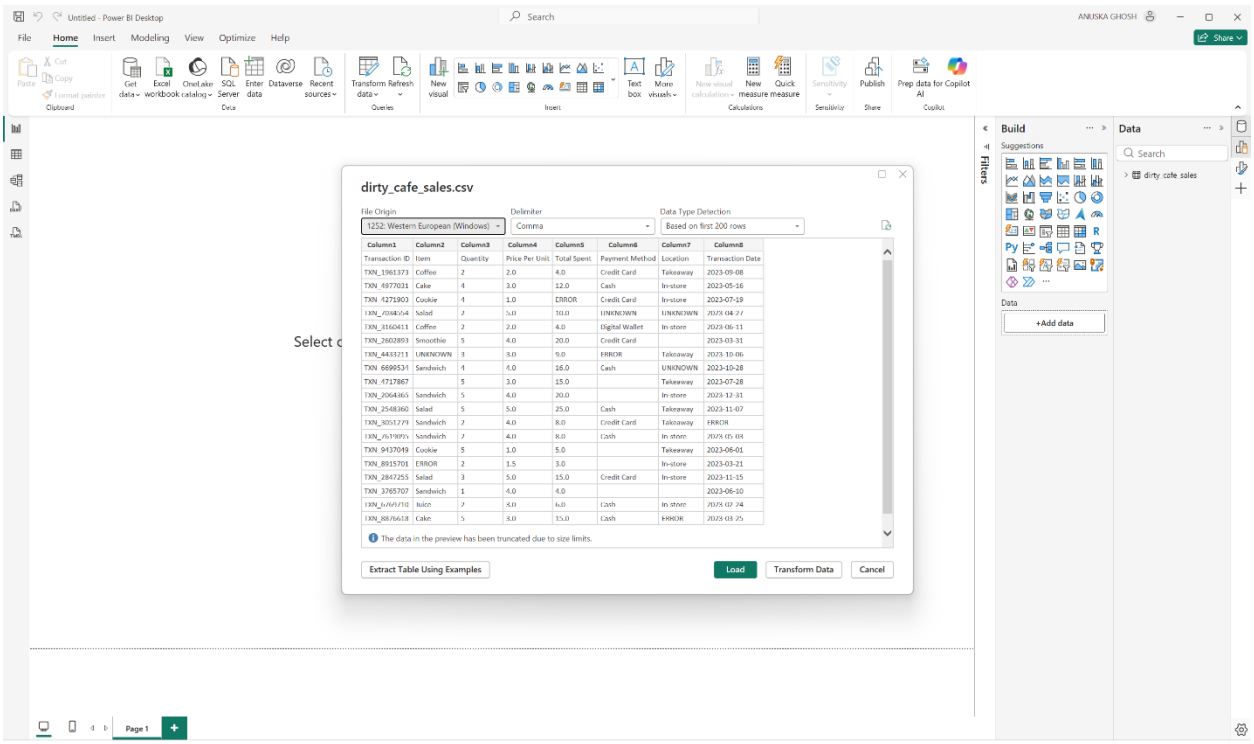


Figure 2: Get Data

Step 3: Data Cleaning in Power Query Editor

The following cleaning operations were performed:

3.1 Promoted Headers

The first row was promoted as column headers using "Use First Row as Headers" option.

3.2 Data Type Corrections

The following columns were changed to appropriate data types:

- **Quantity:** Text → Whole Number
- **Price Per Unit:** Text → Fixed Decimal Number
- **Total Spent:** Text → Fixed Decimal Number
- **Transaction Date:** Text → Date

3.3 Handling Invalid Values

The following invalid entries were identified and removed from the **Item** column:

- **UNKNOWN values** - removed
- **ERROR values** - removed
- **Blank values** - removed

3.4 Removing Incomplete Records

Rows with missing or null values were removed from the following columns:

- Quantity
- Price Per Unit
- Total Spent
- Transaction Date

3.5 Standardizing Inconsistent Entries

Payment Method and Location columns contained inconsistent entries (ERROR, UNKNOWN, blank). These were replaced with "UNKNOWN" to maintain consistency while preserving the rows for analysis.

3.6 Removing Duplicate Records

The Transaction ID column was checked for duplicate records using the "Remove Duplicates" option. No duplicate records were found, confirming that all 10,000 Transaction IDs were unique.

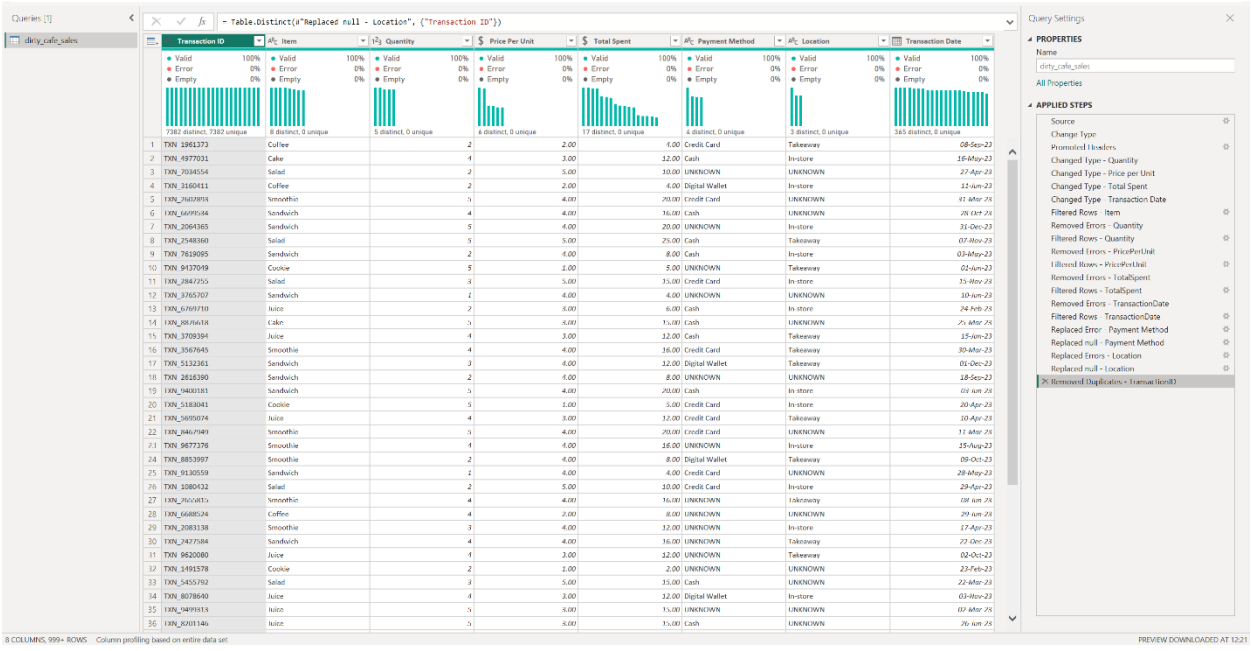


Figure 3: Power Query Editor - Applied Steps + Cleaned Data

Output:

The data cleaning process was successfully completed using Power Query Editor in Power BI Desktop. The following results were observed:

- **Initial Records:** 10,000 rows
- **Final Records after Cleaning:** 7,382 rows
- **Records Removed:** 2,618 rows (invalid, incomplete, or unknown entries)
- **Final Columns:** 8

Column-wise Cleaning Summary:

- **Transaction ID:** 100% Valid. No errors or empty values.
- **Item:** UNKNOWN, ERROR, and blank values removed. Only valid cafe items retained (Coffee, Tea, Juice, Cake, Cookie, Sandwich, Salad, Smoothie).
- **Quantity:** UNKNOWN, ERROR, and blank values removed. Data type changed to Whole Number.
- **Price Per Unit:** UNKNOWN, ERROR, and blank values removed. Data type changed to Fixed Decimal Number.
- **Total Spent:** UNKNOWN, blank, and ERROR values removed. Data type changed to Fixed Decimal Number.
- **Payment Method:** ERROR, blank, and null values replaced with "UNKNOWN". Valid entries retained (Cash, Credit Card, Digital Wallet).
- **Location:** ERROR, Unknown, and blank values replaced with "UNKNOWN". Valid entries retained (In-store, Takeaway).
- **Transaction Date:** Null and error values removed. Data type changed to Date format.

Post-cleaning, all 8 columns show 100% Valid, 0% Error, and 0% Empty values as verified through Column Quality view in Power Query Editor.

Conclusion:

This experiment successfully demonstrated the use of Power Query Editor in Microsoft Power BI to clean and prepare a real-world dirty dataset for analysis. The Cafe Sales dataset containing 10,000 records was imported, inspected, and cleaned through a series of systematic transformation steps. The cleaning process involved promoting headers, correcting data types, removing invalid entries (UNKNOWN, ERROR, blank, null), handling missing values, standardizing inconsistent entries, and removing duplicate records. As a result, the dataset was reduced from 10,000 to 7,382 clean and consistent records, ready for further business analysis. The experiment highlighted the importance of data cleaning as a critical step in the data analytics pipeline, ensuring that business decisions are based on accurate and reliable data.

Business Insights:

The following key business insights were derived from the cleaned Cafe Sales dataset:

- **Data Quality:** The raw dataset contained significant quality issues with 2,618 invalid or incomplete records out of 10,000, representing approximately 26% data quality issues that could have led to inaccurate business decisions if left uncleaned.

- **Item Performance:** The cleaned dataset retains 8 valid cafe items like Coffee, Tea, Juice, Cake, Cookie, Sandwich, Salad, and Smoothie, enabling accurate product-wise sales analysis.

- **Payment Method Insights:** A significant portion of Payment Method data was missing or inconsistent, indicating a need for better data entry practices at the point of sale.

- **Location Data:** A large number of Location entries were missing or invalid, suggesting improvements are needed in capturing customer location (In-store vs Takeaway) during transactions.

- **Data Readiness:** After cleaning, all 7,382 records are complete and consistent across all 8 columns, making the dataset ready for accurate business analysis and dashboard creation.

- **Duplicate Check:** Verification of Transaction IDs confirmed that all records were unique with zero duplicates, ensuring data integrity was maintained throughout the dataset.